Project 2:

Public Portfolio Post

Josh Inacio (This is temporary until I can fix my website already talked to Aileen)

Introduction to the Problem

While deciding what to focus on for this project, I became interested in how machine learning might predict natural disasters. This curiosity stemmed from watching YouTube videos about past events over the last month. Of all the videos I watched, the one that resonated with me the most was about the 2004 Indian Ocean earthquake. This single event claimed the lives of more than 200,000 people and ultimately inspired me to concentrate on studying earthquakes and their impact on tsunamis. At the time, scientists, engineers, and emergency managers wanted to determine whether a tsunami could be predicted from seismic data alone, which is the question I set out to explore.

My research question is: "Given what we know about the factors of an earthquake, like magnitude, depth, and location, are we able to predict whether a tsunami will be triggered or not?" My question connects data science to public safety, a key field that is increasingly in need of tools that can support saving lives. This project not only applies modern machine learning techniques to historical earthquake records but also uses these patterns and factors to reliably flag tsunami events early enough to inform emergency warning systems. With these emergency warning systems informed, lives can be saved and first responders can be deployed early.

Introducing the data

For this project, I worked with the global Earthquake–Tsunami Risk Assessment Dataset, a publicly available Kaggle dataset with 782 significant earthquakes of magnitude 6.5 and above. This data is recorded worldwide between 2001 and 2022, representing a wide time range. Each one of these earthquakes is described by 12 measured features in the form of variables. These include the earthquake's magnitude on the Richter scale, how deep underground the earthquake originated, where it originated in terms of latitude and longitude, and how strongly it was felt by people and by the instruments that record this data. The rest of the data consists of data-quality indicators from seismic monitoring networks.

One of the best things about this dataset is that each earthquake is labeled for whether it produced a tsunami or not, shown through a Boolean variable named "tsunami." Out of the 782 earthquakes recorded in this dataset, 304 triggered a tsunami while the other 478 did not. This comes down to a ratio of 39% triggered vs 61% not triggered. With all things considered, this relatively balanced split made it very well suited for binary classification, which allows machine learning models to sort events into two different categories.

My findings

I trained three different machine learning models on this data and compared their ability to correctly identify tsunami-generating earthquakes. In my findings, gradient boosting had the highest performance, with an accuracy of 70.1% and an AUC of 76.9%, followed closely by the random forest model with 65.6% accuracy and an AUC of 73.9%, and finally logistic regression with 58.6% accuracy and an AUC of 49.7%

Because gradient boosting was the most accurate model, I used it for my feature importance analysis. The most surprising insight came from this analysis. My initial hypothesis was that magnitude would be the most important feature, but out of all the predictors I focused on, the one that held the most weight was geographic location

- **Geographic location (longitude + latitude) accounted for 82.8% of predictive importance**
- **Depth contributed ~11.9%**
- **Magnitude contributed only ~5.3%**

Geographic location is connected to tectonic plate boundary location, which I now see as one of the strongest predictors of tsunami generation. This also makes sense when you take into account that, according to NOAA, "A tsunami is generated when a large mass of Earth on the bottom of the ocean drops or rises, thereby displacing the column of water directly above it. This type of displacement commonly occurs in large subduction zones, where the collision of two tectonic plates causes the oceanic plate to dip beneath the continental plate to form deep ocean trenches."

Depth also still matters, as shallow earthquakes are more likely to displace the ocean floor and generate waves

The key takeaway: while both depth and magnitude play a role in predicting whether a tsunami is triggered or not, geographic location dominates, and from this data I can confidently say that machine learning models can reliably predict tsunami occurrence from seismic features, especially when using gradient boosting.

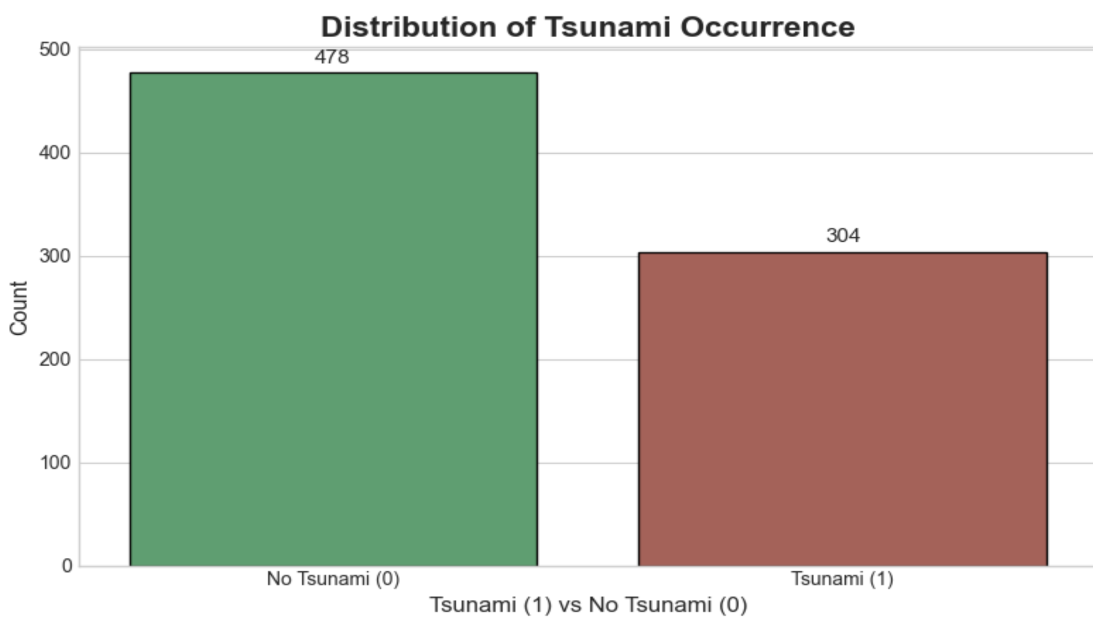
What I Learned

Working on this project taught me the power of machine learning, especially when studying earthquakes and tsunamis. I learned how to take a real scientific dataset, clean it, and turn it into a model that could potentially support public-safety decisions. Comparing Gradient Boosting, Random Forest, and logistic regression helped me understand the effectiveness of each model and the trade-offs between different algorithms on nonlinear problems.

I also realized that my initial hypothesis will not always be correct. I originally expected magnitude to be the main driver of tsunami risk, but the models showed that this was not the case and that geographic location and depth were the main predictors. This pushed me to read more about plate boundaries and connect my model's feature importance back to real-world settings described by NOAA.

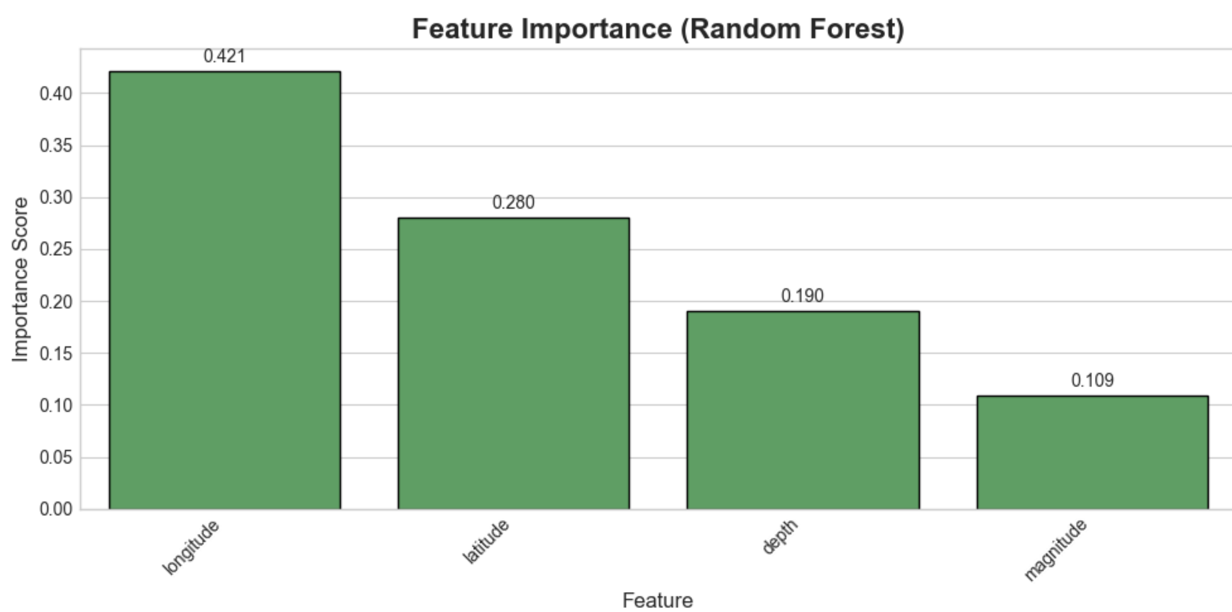
Finally, I not only gained experience communicating technical results to non-experts in a way they can understand, but I also moved beyond just reporting accuracy and AUC to explain why location dominates. I also had to convey how early warning systems could benefit from this model and what the limitations of this project are. Overall, I had a lot of fun coding this project and gained many insights into machine learning and its applications in an environmental context.

Visualizations/ Appendix

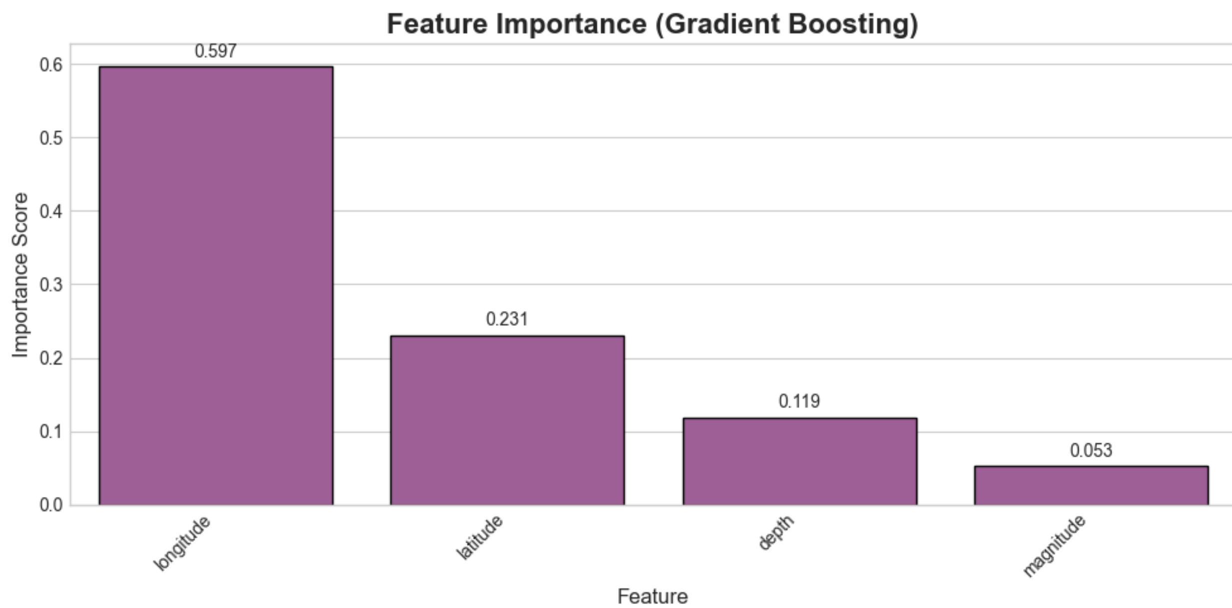


This shows the 69/31 ratio split between No tsunami and Tsunami's that is shown in the dataset. A model that always predicts no tsunami would already have a 69% chance of being right so this helped me discover that accuracy is not the only parameter that I should use for this dataset.

This feature-importance plot originates from our logistic regression model, our least accurate out of the three. This models still provides a first look into what inputs our models relies on the most and gave us an initial hypothesis and about the conclusions found in my findings section. These results were later compared with our better models.



This feature-importance for our Random Forest model, the second most effective model shows that longitude and latitude are our most influential predictors and following that its depth and magnitude. From this conclusion, we are able to further solidify that our features are consistent across models which further strengthens the patterns that we observe.



The last and most effective of our feature importance models was Gradient Boosting, which achieved our highest overall performance. Its reliance on location variables highlights the dominant predictor in tsunami occurrence. Reinforcing the patterns seen in both logistic regression and Random Forest. Because this was our best model we relied heavily on this model for our primary evidence when drawing our final conclusions.

Resources:

"Introduction to Global Historical Tsunami Data." *National Centers for Environmental Information*, National Oceanic and Atmospheric Administration, https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis/global-historical-data.

Zaki, A. M. (n.d.). Global Earthquake–Tsunami Risk Assessment Dataset [Data set]. Kaggle. https://www.kaggle.com/datasets/ahmeduzaki/global-earthquake-tsunami-risk-assessment-dataset

AI Acknowledgement

Perplexity was used to troubleshoot errors and warning messages, and correct grammatical errors.